IBScovars: reference-paper-style application

Simultaneous statistical inferences of treatment–control mean differences

Methods / dose comparison	Point estimate (treatment − control)	95% simultaneous confidence interval
Proposed method with shrinkage		
1 vs. 0	0.243	(-0.121, 0.608)
2 vs. 0	0.352	(-0.013, 0.718)
3 vs. 0	0.193	(-0.127, 0.514)
4 vs. 0	0.298	(-0.087, 0.682)
Proposed method without shrinkage		
1 vs. 0	0.243	(-0.122, 0.609)
2 vs. 0	0.352	(-0.005, 0.710)
3 vs. 0	0.193	(-0.121, 0.507)
4 vs. 0	0.298	(-0.093, 0.689)
Dunnett's procedure with naive case deletion		
1 vs. 0	0.292	(-0.067, 0.651)
2 vs. 0	0.326	(-0.010, 0.662)
3 vs. 0	0.208	(-0.137, 0.554)
4 vs. 0	0.305	(-0.070, 0.679)

One application analysis; 40 propensity-score imputations as specified by the uploaded script. The target missingness is 30% MAR; the realized missingness is 118/369 = 31.98%.

Shrinkage and non-shrinkage share the same 40 imputations and point estimates. Naive case deletion is the comparator explicitly included in the supplied script. No additional simulation loop or imputation-method comparison was added.

The accompanying forest plot reproduces the script’s reference-paper layout: 40 individual completed-dataset results, both pooled procedures, and naive case deletion, separately for each dose contrast.

Forest plot showing the point estimates and 95% confidence intervals

IBScovars, 30% MAR; propensity−score imputation; individual imputations, pooled results, and naive case deletion

